Unit 6: Microcontrollers — Week 1 Baseline Assessment

Section 1: Student Information & Workplace Context

Welcome to the engineering development team. This baseline diagnostic helps establish your current strengths and where to focus your development throughout Unit 6. It is not a graded pass/fail exam.

1. Full Name

2. Student ID / College Number

3. Have you studied electronics before?

- ◯ No previous experience
- ◯ Some experience
- ◯ Regular experience
- ◯ Significant experience

4. Have you built electronic circuits before?

- ◯ Never
- ◯ Once or twice
- ◯ Occasionally
- ◯ Frequently

5. Have you used a multimeter before?

- ◯ Never
- ◯ With support
- ◯ Independently
- ◯ Confidently

6. Have you written program code before?

- ◯ Never
- ◯ A little (block-based / Scratch)
- ◯ Text-based (Python / C / C++ / Arduino)
- ◯ Regular programming experience

Section 2: Engineering Toolbox Diagnostic

Complete the questions independently. If you are unsure, record your best reasoning.

7. Component Matching: Match each component to its function (A. Resistor, B. LED, C. Capacitor, D. Switch)

Functions: 1. Emits light | 2. Stores charge | 3. Limits current | 4. Opens/closes circuit

Format answer as A-3, B-1, C-2, D-4:

8. Which of the following components have polarity (+ / -)? (Select all that apply)

- ☐ Resistor
- ☐ LED
- ☐ Electrolytic Capacitor
- ☐ Battery / DC Supply
- ☐ Switch

9. Briefly explain what could happen if a polarised component is connected backwards in a circuit.

10. Ohm's Law Calculation: A 12 V supply is connected across a 240 Ω resistor. Calculate the circuit current in Amperes (I = V / R).

11. Unit Conversion: Convert 4.7 kΩ to Ω, and convert 25 mA to A.

Section 3: Digital Systems Baseline

12. What is the difference between an analogue signal and a digital signal?

13. Binary Conversion: Convert the 8-bit binary value 00001010 to decimal.

14. In a 5 V microcontroller system (e.g. Arduino Uno), what voltage levels correspond to Logic HIGH (1) and Logic LOW (0)?

15. Logic Gates: Complete the output for an AND gate and an OR gate when Input A = 1 and Input B = 0.

Section 4: Programming Baseline

16. Code Tracing: Review the C++ code snippet below. What exact message will this program display?

```
int temperature = 25;
int limit = 30;
if (temperature > limit) {
    cout << "WARNING";
} else {
    cout << "NORMAL";
}
```

17. Variable Assignment: If count = 3 and then count = count + 2, what is the final value stored in count?

18. Loop Tracing: In a loop for (int i = 0; i < 4; i++), how many times does the loop execute?

19. Code Modification: State what line you would change in the temperature program above if you wanted the warning to trigger at 20 degrees instead of 30 degrees.

Section 5: Practical Circuit, Pre-Power Safety & Evidence Upload

20. Pre-Power Safety Gate Confirmation (Confirm each check completed before turning on power):

- ☐ Circuit layout matches schematic
- ☐ Resistor value confirmed (330 Ω)
- ☐ LED polarity verified (Anode to +5V via resistor, Cathode to 0V)
- ☐ Power supply set to 5V DC before connecting
- ☐ Meter configured properly for measurement

21. Recorded Multimeter Measurements (Supply Voltage, Resistor Voltage, LED Forward Voltage, Circuit Current):

22. Practical Evidence Upload (Upload your Wokwi Simulation screenshot, Breadboard Photo, and Multimeter Display):

File Upload (Max 3 files: Image, PDF, DOCX)

23. Individual Baseline Self-Assessment (Rate your current confidence 1–5):

Topic / Skill	1 (Need Support)	2	3	4	5 (Fully Confident)
Electronics & Components	○	○	○	○	○
Digital Systems & Binary	○	○	○	○	○
Microcontroller Programming	○	○	○	○	○
Electrical Measurement & Multimeters	○	○	○	○	○
Circuit Construction on Breadboard	○	○	○	○	○

24. Personal Development Priority: Based on today's baseline, what is the #1 skill you want to develop during Unit 6?

25. Exit Ticket Reflection: List three things you did confidently, and one thing you found challenging today.